



Meta-learning-Based Model to Select Algorithm for Capacitated Vehicle Routing Problem

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Abstract

The Capacitated Vehicle Routing challenge is a typical hard combinatorial optimization problem for which numerous algorithmic techniques have been developed. Empirical studies have consistently shown that there is no universally accepted optimal strategy that works equally well on all instances of a problem. It indicates that performance of an algorithm varies across different scenarios. Majorly, performance of algorithms depends on structural and other properties of problem instance. The challenge of identifying the most effective solution method from a predefined set of options is addressed through algorithm selection problem. To this end, a meta-learning-based categorization system is proposed. This study employs a heuristic algorithm to a collection of instances, categorising them as either ‘solved’ or ‘not solved’, depending on whether the algorithm successfully solves the instance inside specific configuration. The label created along with the characteristics of instance provides a meta example in the meta-data. A set of binary classifiers are learned to predict the suitability of algorithm for different instances. Experimentation is done on 5000 CVRP instances. Eight different classifiers demonstrate accuracy and other performance metrics up to 90%.

Keywords Vehicle routing problem · Heuristic algorithm · Machine learning · Meta-learning · Algorithm selection problem

Introduction

The Capacitated Vehicle Routing Problem (CVRP) was proposed by Dantzig and Rasmer [8] in 1959, is a basic combinatorial optimization problem with many applications in logistics and networking sectors. It aims to build low-cost routes to supply consumer requests utilizing the available fleet of fixed capacity vehicles. It is a frequently researched topic, not only because of its complexity, but also because CVRP and its variants represent various real-world problems such as garbage collection, transportation of people and products, school bus routing, etc. Since CVRP is an NP hard [14] issue, exact approaches are impractical for large-scale

cases. Therefore, it has received a great deal of attention from the algorithm design and optimization community. Heuristic [23] and meta-heuristic (MH) approaches [12] provide near-optimal solutions in modest computing time. Heuristic algorithms are deterministic and operate within problem space, while MHs are stochastic and function in solution space. Search for optimal solutions can commence with either a single solution (trajectory-based) or a population of solutions. Adjustments are being made to heuristics to tailor them for specific variants. Latest updates on MH for optimization problem provided in [5, 20]. In addition to solution cost, carbon emission, fuel consumption, customer satisfaction, driver working hours are other major concerns. Electric and traditional vehicles have different pollution emissions and charging routes, so a reasonable mix of the two will reduce carbon emissions during distribution and boost logistics system profits [9].

In recent years, researchers integrate machine learning (ML) with exact and heuristic algorithms to solve Rich VRP, defined as multi objective VRP. Set partitioning (an exact approach) [3] is combined with reinforcement learning [11] and enhanced Q-learning is applied with Tabu

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search in [1] to address VRP. A systematic overview of ML methods applied to VRPs is provided by Bogrybayeva et al. [4]. The No-Free-Lunch Theorem [26] states that all these approaches produce satisfactory outcomes but perform equally well on an average. As a result, research into methods for selecting the best strategy has gained attention.

One approach to this challenge is through algorithm selection problem (ASP) [18]. The process of determining the best from given methods for an instance based on its characteristics is defined as ASP. ML community approached the ASP through meta-learning. The objective is to get a more profound comprehension of correlation between problem description and algorithm performance. It is used in optimization, data mining, sorting, scheduling, cryptography, where there exist numerous algorithmic techniques to solve the problem. Main emphasis of this work is to address CVRP inside the meta-learning framework. Instance characteristics, known as meta-features, were extracted from the problem definition. Each instance is then labelled with algorithm performance metric. A predictive model is then learned and can be used thereafter.

Binary classification is an ML task which categorize examples into two classes. It is the foundation of multi-class and multi-label classification. In the proposed work, Clarke and Wright saving heuristic algorithm [7] is applied to a set of 5000 CVRP instances [10] and based on whether it can solve the instance or not within a certain configuration, each instance is labelled as ‘solved’ or ‘not solved’. Training dataset is prepared using twenty-three features described by Steinhaus [21] and labels generated. Hence meta-learning problem is framed as binary classification. Classifiers learned demonstrated accuracy up to 90%.

In rest of the paper, concise overview of meta-learning for algorithm selection is outlined in “Meta-learning for Algorithm Selection” section. In “Related Work” section, previous work on algorithm selection for CVRP is presented. Experiment performed and its findings are discussed in “Proposed Work” and “Results” sections, followed by conclusions derived and future directions in “Conclusion and Future Work” section.

Meta-learning for Algorithm Selection

Meta-learning is defined as inductive learning carried out on algorithm performance data that is obtained upon previous executions of algorithms on problem set. It aims to exploit algorithms performance data either to enhance the performance or choice of algorithms [6]. It is of great interest to gain the insights of algorithm performance to achieve better algorithm design [13]. Knowing the association between problem attributes and performance of techniques assist end

users in choosing the solution approach which will perform better than other available options. It provides a framework which enables to learn a model on extracted metadata that leads to automated algorithm selection. Meta-learning was initially applied to classification and clustering. Later, generalized for non-learning tasks like planning, searching, sorting, forecasting, constraint satisfaction, optimization, bioinformatics, cryptography, etc. [19]. It has been applied successfully to automated selection, performance analysis, run-time prediction, hyper-parameter tuning, portfolio design of algorithms. Like any ML task, meta-learning is also performed in phases as illustrated in Fig. 1.

- (i) Metadata collection: In this phase, meta-features are extracted from instances and meta-labels are generated by applying algorithm on problem set. The combination of meta-features and meta-labels creates a meta-database that is used for learning purposes. Meta-features that impact the performance of algorithms and measurable in polynomial time need to be extracted from the problem instances.
- (ii) Training: This phase focuses on comprehending the meta-data to establish the mapping from problem characteristics to performance measures of the solution methods using ML algorithms.
- (iii) Prediction: The application of a learning algorithm results in the development of a model, which enables the system to operate in advising mode.
- (iv) Maintenance: Continuous adaptations of model obtained using feedback and further analysis.

Both statistical and ML approaches have been used successfully. Statistical learning emphasizes model interpretability, whereas ML prioritizes predictability.

Related Work

Towards automated algorithm selection for VRP, Nygard et al. [15] have made a significant contribution. Authors asserted that performance of various heuristic algorithms for VRP is sensitive to spatial distribution of clients. A complex, multi-layered neural network is created to choose the best heuristic method for a particular VRP instance. The problem characteristics that were extracted had a close relationship with the clusters of nodes that made up the problem. In contrast, a simpler artificial neural network (ANN) designed by Tuzun et al. [24] for choosing the best VRP method among Clarke and Wright saving heuristic, Gillett and Miller sweep algorithm and Koskosidis and Powell capacitated clustering algorithm. The researchers established an experimental framework and produced 1280

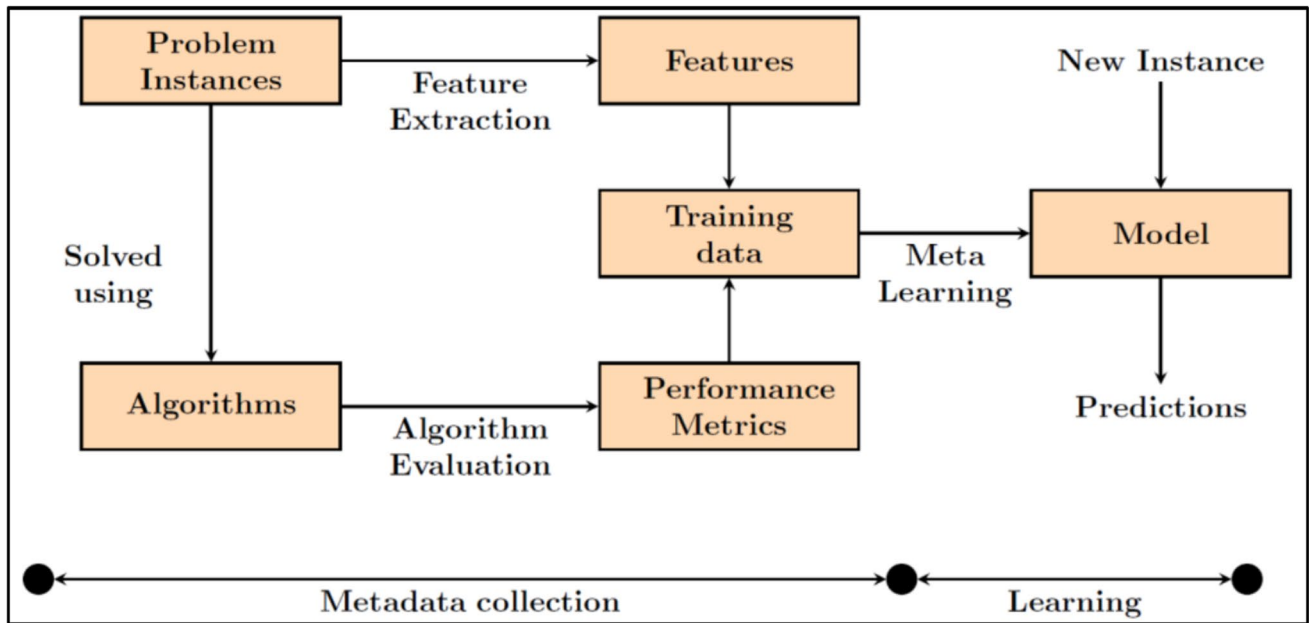


Fig. 1 Meta-learning process

CVRP instances by controlling factors related to clustering, node spread, central location, tour lengths, and demand. All of these features, were found statistically significant in influencing solution cost of a VRP algorithm. ANN were trained for both recommending best heuristic method and predicting average tour length.

Till there were no reference to Algorithm Selection Problem (ASP) formalized by J Rice [18]. Algorithm selection involves selecting or recommending the most suitable technique for a specific instance from various potential solutions available for a given problem, based on its attributes. Based on ASP model, Steinhaus [21] framed a classification problem, taking instances from CVRPLIB¹ that were solved using three techniques: Clarke and Wright, sweep algorithm and self-organizing maps (SOM). Author characterized each instance with 23 features based on demand, spatial distribution of clients and fleet of vehicles. She labelled them with the algorithm that produced solution closest to optimal or best known one. SOM technique was used to learn a classification model for automated algorithm selection. She observed that, in contrast to unsupervised SOM, which predicts correctly the best algorithm 59% of the time, supervised SOM does 74% of the time. Instead of experimenting on instances synthesized randomly, Uchoa et al. [25] generated 100 instances such that instance characteristics follow the desired distribution. Later on, following the same idea, Fellers [10] fabricated 5000 CVRP instances and extended work of Steinhaus [21] with same feature set and four algorithms. Three models KNN, Decision Tree, and

Single-Layer Perceptron learned for predicting best algorithm. The reported accuracy for each is 76.4%, 75.1% and 79.4% respectively. Rasku et al. [17] extracted 433 features [16] from 454 CVRP instances from the CVRPLIB. Each instance was submitted to fifteen heuristics and labelled with the best heuristic based on optimality gap and lapsed time data. Authors also applied dimensionality reduction on the feature data and removed the sequential savings algorithm from algorithm space since it didn't get labelled as best for any of the instance. Performance patterns were learnt using four classifiers: multilayer perceptron, support vector machine, random forest, 3-nearest neighbour. They claimed that classification was largely aided by attributes generated using local search techniques, tightness, depot location, and node distribution. Asin et al. [2] considered a portfolio of fast heuristic algorithms and used the meta-learning framework incorporating both algorithm selection and configuration. Thus, availability of numerous solution techniques demands careful investigation of instances. Meta-learning process undertaken for the purpose is discussed in next section.

Proposed Work

In this work, a set of 5000 CVRP instances [10] is used, in which instances are deliberately designed to ensure diversity in spatial distribution of nodes, depo positions, demands etc. This was not possible with instances that are generated at random. Each instance carries information about number of nodes, Euclidean coordinates, demands and vehicle

¹ <http://vrp.atd-lab.inf.puc-rio.br/index.php/en/>

capacity. Nodes are distributed in a grid of 500×500 following a pattern either equidistant, random, clustered or random-clustered. Depot node is located either at centre, one of the corners, or random position in the grid. Demands for the nodes also follow a pattern. Utilizing the Google OR-Tools² library, the Clarke and Wright saving heuristic [7] is applied to all instances. It first creates separate routes for every node, then repeatedly merge them provided capacity constraint is not violated and cost of route increased marginally. The execution time of the heuristic is 30 s, while being constrained by the capacity and number of vehicles. Equation 1 determines the number of vehicles for each instance where d_k is demand of k th node and Q is the vehicle capacity.

$$\text{Number of vehicles} = \frac{\sum_{k=1}^n d_k}{Q} \quad (1)$$

Each example is designated with the label ‘Y’ if a solution is obtained, and ‘N’ otherwise. Each instance is represented

Table 1 Classification algorithms

Sn	Classification algorithm	Remarks	Test accuracy	F1_score
1	Logistic Regression	Simplest algorithm for binary classification	0.785	0.762
2	Linear SVM	Find the best line or decision boundary, called the hyperplane for linearly separable data	0.781	0.751
3	Radial SVM	Used for nonlinearly separable data	0.785	0.755
4	Decision Tree	In a tree-structured model, core nodes, branches, and leaf nodes reflect dataset properties, decision rules, and outcomes respectively	0.868	0.867
5	Random Forest	Ensemble learning integrates various weak classifiers to address a complex issue or to enhance performance	0.880	0.874
6	ADA Boost	Iteratively train weak classifiers, each successive classifier emphasizes on examples that were misclassified	0.851	0.845
7	Gradient Boosting	Combines multiple weak learners to build a strong learner, and gradually increases its accuracy	0.893	0.889
8	KNN	Outcome of K nearest points in the training dataset is used to predict the category of a new data point	0.774	0.757

using twenty-three features. A binary classification problem is so constructed and learned using eight different classifiers [22], listed in Table 1 with their default configuration available in scikit-learn³ library. Every algorithm used a 70/30 split for training and testing. Classifiers are evaluated and reported in next section.

Results

In this section, eight different classifiers are learned and their performance metrics are reported. Accuracy and F-score of all classifiers are presented in Table 1. Among all, GradientBoosting algorithm provides highest accuracy and F1 score. Out of 1500 test instances 1328 are correctly classified. Classification report confirms the precision, recall, F1 score, and support scores for both the classes. DecisionTree algorithm learned a tree of depth 18 presented the accuracy score of 0.868. For visualization purposes top two splits of resultant tree is shown in Fig. 2. It allows us swiftly examine the process of categorization.

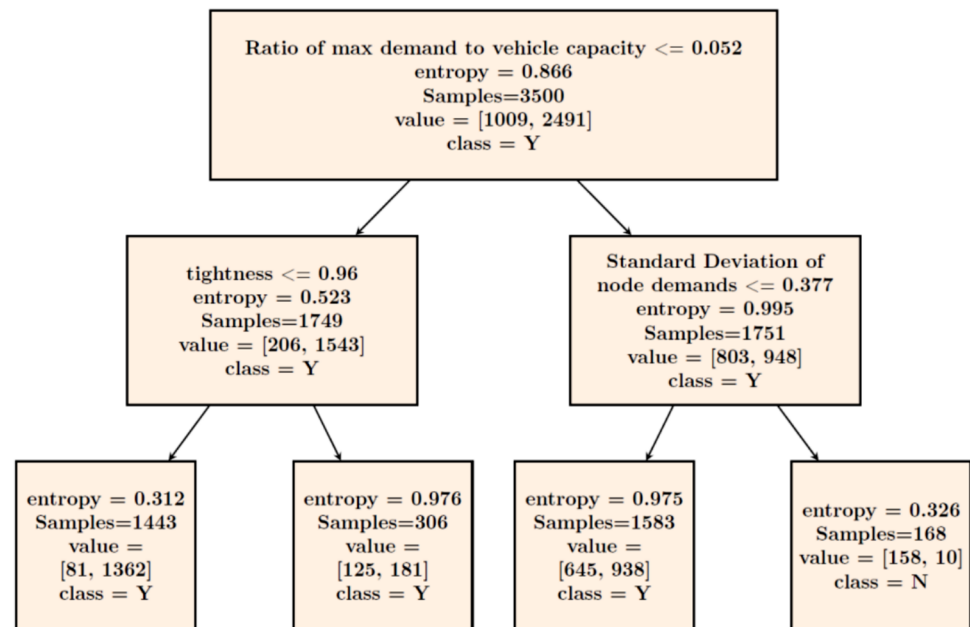
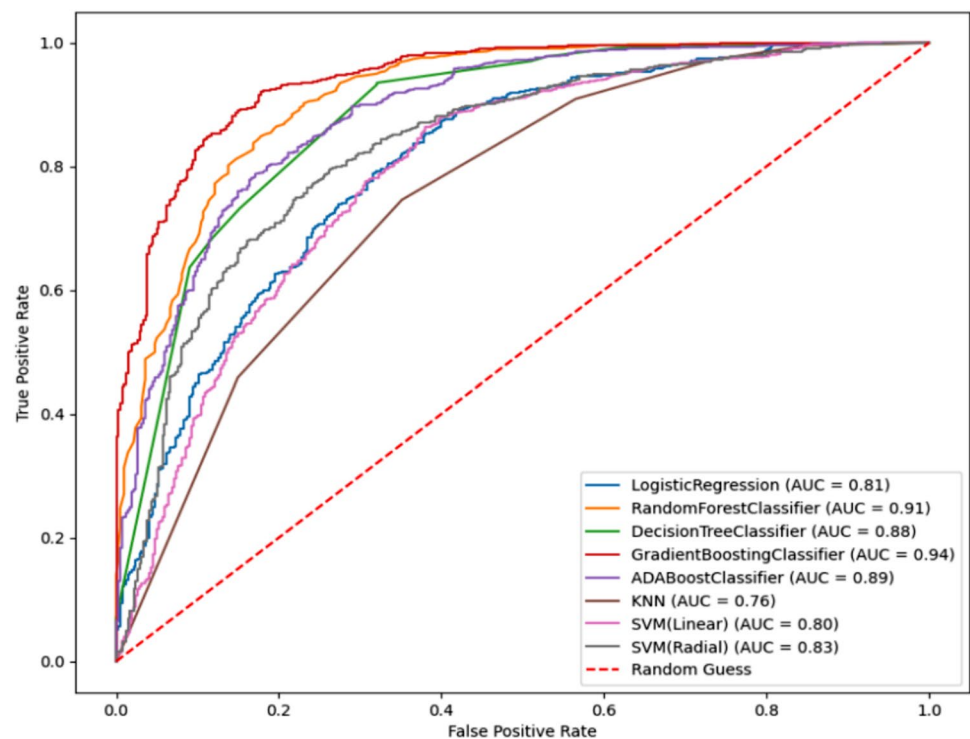
Optimal performance is achieved by hyper-parameter tuning using Grid Search CV with tenfold cross-validation. The optimal parameter values obtained for the criteria, Max_depth, Min_sample_split, and Min_samples_leaf are ‘entropy’, 8, 7, and 1, respectively. ROC and AUC scores provide yet another means of assessing performance. Figure 3 illustrates ROC Curve for all classifiers learned along with AUC score.

Conclusion and Future Work

In this work, a meta-learning-based model has been developed to check suitability of a heuristic algorithm for CVRP instances. To accomplish this task, a classification dataset was constructed by considering twenty-three meta-features, one heuristic approach, and a substantial number of CVRP instances. Exhaustive experimentation has been performed on generated meta-dataset by eight classification algorithms. Results were computed and visualized for detailed analysis. We achieved encouraging results. GradientBoosting shows the highest accuracy, F1 score, and AUC score whereas DecisionTree model allows us to swiftly examine the categorization process. Performance can further be enhanced by optimizing hyper-parameters, augmenting data quality and quantity, and employing advanced deep-learning methodologies. Future work will also involve experiments with larger instance sets, more heuristics and meta-heuristics, considering the algorithm selection as a multi-class or multi-label

² <https://developers.google.com/optimization/routing/vrp>

³ <https://scikit-learn.org/stable/>

Fig. 2 Decision tree output of DecisionTreeClassifier**Fig. 3** ROC curve for classification algorithms

classification problem. The scope of future study will not be restricted to CVRP. The VRP and its variants will be investigated further within the framework of algorithm selection problem for the foreseeable future.

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Declarations

Competing interests The authors declare that they have no competing interests.

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